

Where do cosine frequencies fold?

Background

For a zero-phase cosine sampled at f_s , frequencies separated by an integer multiple of f_s are indistinguishable, and cosine is unchanged by reversing the sign of frequency. The baseband alias is the unique equivalent nonnegative frequency in $[0, f_s/2]$. This folding rule applies here to zero-phase cosines only.

Your task

Find the baseband alias of each supplied frequency. Sample each original cosine and its alias, then report the maximum sample difference for each pair. Display the supplied frequency comparison plot.

Function contract and assumptions

`fold_frequencies(freqs, fs)` returns aliases with the same shape as the one-dimensional input.
`verify_aliases(freqs, aliases, fs, N)` returns one maximum absolute sample error per pair, using $n=0, \dots, N-1$. Frequencies are nonnegative, $f_s > 0$, and $N \geq 1$.

Example / correctness check

For `freqs=[0,3,13,17,23,40]` and `fs=20`, `aliases=[0,3,7,3,3,0]`. With `N=20`, all verification errors should be below $1e-10$. Include the $f_s/2$ boundary correctly.

Exam instructions

Complete only the TODO functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF pages 5, 19-22; cosine folding is explained above.